Data-Level vs. Algorithm-Level Solutions for Imbalanced Classification: A Comparative Study Using Logistic Regression and Random Forest

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Abstract
Imbalanced datasets present a significant challenge in supervised machine learning because one class often contains substantially more examples than another. As a result, machine learning models may achieve high overall accuracy while performing poorly on the minority class, which is often the class of greatest interest. This project compares several commonly used data-level and algorithm-level techniques for handling class imbalance using two publicly available datasets: Credit Card Fraud Detection and Breast Cancer Wisconsin. Logistic Regression and Random Forest were evaluated using a baseline model together with Random Oversampling, Random Undersampling, SMOTE, and Class Weighting. Performance was measured using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, and ROC curves. The results showed that no single imbalance-handling strategy consistently produced the best performance across both datasets. Instead, the most effective approach depended on the dataset, the learning algorithm, and the evaluation metric being considered.

1. Introduction
Many real-world machine learning applications require models to identify rare but important events. Examples include fraudulent financial transactions, disease diagnosis, spam detection, and cybersecurity. In these problems, the cases that matter most often represent only a small portion of the available data. This creates an imbalanced classification problem where machine learning models tend to favor the majority class while struggling to correctly identify the minority class.

To address this challenge, researchers have developed a variety of techniques for improving classification performance on imbalanced datasets. These techniques generally fall into two categories. Data-level approaches modify the training data before model training, while algorithm-level approaches adjust how the learning algorithm handles class imbalance. Choosing the most appropriate approach depends on the characteristics of the dataset and the machine learning model.

This project evaluates several commonly used imbalance-handling techniques using two datasets with different levels of class imbalance. The Credit Card Fraud Detection dataset represents a highly imbalanced classification problem, while the Breast Cancer Wisconsin dataset contains a more moderate imbalance. Rather than artificially changing the class distribution, this study compares how these techniques perform on two real-world datasets that naturally exhibit different imbalance characteristics.

2. Research Question
The main objective of this project is to answer the following research question:

> ***Which imbalance-handling strategy produces the best classification performance across datasets with different levels of class imbalance?***

To answer this question, Logistic Regression and Random Forest were evaluated using multiple imbalance-handling techniques. Model performance was compared across both datasets using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, and ROC curves.

3. Methodology
This project used two supervised learning algorithms: Logistic Regression and Random Forest. Logistic Regression was selected because it is a widely used baseline classifier for binary classification problems, while Random Forest was selected because of its ability to capture more complex relationships through ensemble learning.

- Baseline (no imbalance handling)
- Random Oversampling
- Random Undersampling
- Synthetic Minority Oversampling Technique (SMOTE)
- Class Weighting

The Credit Card Fraud Detection dataset contains more than 280,000 transactions, with fraudulent transactions representing less than one percent of the data. To reduce computation time, the majority class in the training set was randomly limited to 25,000 legitimate transactions while preserving all fraudulent transactions. The 25,000 legitimate transactions were selected randomly using a fixed random seed (random_state = 42) to ensure reproducibility. The Breast Cancer Wisconsin dataset was evaluated using its original class distribution.

Both datasets were divided into training and testing sets using a stratified train-test split so that the class distribution remained consistent across both sets. Logistic Regression used standardized input features before training, while Random Forest followed the same training and evaluation process for consistency. All models used the standard decision threshold of 0.50 when assigning class labels.

Performance was evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, confusion matrices, ROC curves, class distribution plots, and F1 score comparison charts.

4. Results

The results showed that the effectiveness of an imbalance-handling strategy depended on both the dataset and the machine learning algorithm.

For the Credit Card Fraud Detection dataset, Random Forest generally produced the strongest overall performance. The baseline Random Forest model achieved the highest F1 Score, while Random Undersampling achieved the highest ROC-AUC among the evaluated Random Forest models.

For the Breast Cancer Wisconsin dataset, nearly every model achieved strong classification performance. Because the dataset is smaller and less severely imbalanced than the fraud dataset, the differences between imbalance-handling techniques were less noticeable. Several models achieved ROC-AUC values close to 1.0, while small differences in Precision, Recall, and F1 Score showed that the sampling strategy could still influence overall performance.

The confusion matrices, ROC curves, and F1 score comparison charts also illustrated that the model with the highest ROC-AUC was not always the same model with the highest F1 Score.

5. Discussion

One of the main findings from this project is that there is no single imbalance-handling strategy that consistently performs best across different datasets and evaluation metrics. Although Random Forest generally achieved the strongest overall performance, different sampling techniques produced different trade-offs between Precision and Recall. The Breast Cancer Wisconsin dataset produced consistently strong results across most models because the classes are easier to separate than those in the fraud dataset. Accuracy alone was not enough when evaluating imbalanced datasets; Precision, Recall, F1 Score, and ROC-AUC provided a more complete assessment.

6. Conclusion

This project compared several commonly used data-level and algorithm-level techniques for handling imbalanced classification problems using Logistic Regression and Random Forest. Two datasets with different natural levels of class imbalance were used to evaluate how each approach affected classification performance.

The results showed that no single imbalance-handling strategy consistently outperformed the others across every dataset and evaluation metric. Overall, Random Forest generally achieved the strongest performance, while Logistic Regression provided a useful baseline. Future work could explore additional imbalance-handling techniques, such as Balanced Random Forest or Focal Loss.

7. GitHub Repository

https://github.com/Natnail101/imbalanced-classification-comparison